NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305244
Lead Applicant	Mr Peter Wilding
Research Title	How can we improve inclusive participation in hyperacute and acute stroke research at Hyperacute Stroke Research Centres (HSRCs) in the United Kingdom?
Plain English Summary	**Background** If you have a blocked blood-vessel or bleed in the brain (stroke), you should be taken to a specialised centre for urgent treatment to help prevent death and disability. Such centres are called Hyperacute Stroke Units (HASUs) and those that specialise in research are accredited NIHR Hyperacute Stroke Research Centres (HSRCs). If eligible, you should be offered participation in a clinical trial, this might be because no established-treatment option is available for you or because the research option is potentially safer and more effective. However, this doesn't always happen. Even at centres that specialise in urgent treatment of strokes and hyperacute research, patients do not always get the option of participating in a research treatment. Also, if you are a woman, non-White, from a marginalised community or poorer socio-economic background, you are less likely to participate in research treatments. This means you have less treatment-options available to you when needed most to prevent death or disability. **Aim of the Research** My study aims to find out why and what steps we can take to improve inclusive-participation at HSRCs in the United Kingdom. My study is needed now because the burden of stroke is estimated to increase by 35% by 2050. Currently stroke is the main cause of adult-disability and the third-biggest killer in the UK, costing £26 billion annually. We need more treatment options to help reduce death and disability, and reduce the impact of stroke. Also, equality diversity and inclusion is a national priority that requires urgent action. Patients and carers will benefit from my study because it will widen-access and improve participation in clinical trials for eligible patients at HSRCs across the UK. Inclusive-participation will enhance reliability of the evidence produced. Society will benefit because clinical trials will successfully finish on time or earlier, informing guidelines and making clinical treatments available sooner. This, in turn, will reduce the human and economic burden of stroke nationwide. My study will improve the stroke service and also contribute to national strategies related to equality, diversity and inclusion, embedding research into clinical practice, improving commercial-study performance, and improving access to clot-removal (thrombectomy). **Design and Methods** To investigate the issues and develop a solution, I will use the Design Council's Double Diamond Framework, its design-thinking principles, and Normalisation Process Theory (NPT). NPT is a framework to understand the implementation, embedding, and integration of interventions in healthcare settings. These frameworks have been used in healthcare to achieve successful outcomes in complex interventions. I will adopt a mixed-methods approach: I will speak to patients and carers across different demographics who have experienced stroke to understand what the issues and opportunities in emergency stroke research are. I will observe and speak to stroke service colleagues who deliver research services at 4 HSRCs that have agreed to participate. Following that, in partnership with patients and key stakeholders, I will co-design and test the solution at those centres to improve inclusive participation. **Patient and Public Involvement (PPI)** My stroke PPI-group, composed of patients and carers with lived-experience of stroke, helped and continue to shape my study and identify areas of focus, navigate ethical issues, and validate outcomes. For example, members have confirmed the need for inclusive-participation and to find a sustainable-solution for the post-code lottery access to thrombectomy-research. Also, PPI-input has informed how I am thinking about how and when patients/carers should be contacted, and how gender, ethnicity, and culture can bear on receptivity to research. **Dissemination and Impact** I will share my findings with patients, public roadshows, HSRCs, and wider stroke-community. I will also present at national and international stroke conferences and publish in scientific journals.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
Imperial College of Science, Technology and Medicine

Partner organisation	Imperial College Healthcare NHS Trust

Research Title	How can we improve inclusive participation in hyperacute and acute stroke research at Hyperacute Stroke Research Centres (HSRCs) in the United Kingdom?

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2025

Research Type	Other (please specify)

Please provide more details on your selection above.
Mixed Methods

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
10/2020	01/2023	MSc	Management for Future Leaders	United Kingdom (England)	University of Reading	Distinction
10/2020	04/2024	Level 7 Senior Leader Master's Degree Apprenticeship (SLMDA)	Level 7 Senior Leader Master's Degree Apprenticeship (SLMDA)	United Kingdom (England)	University of Reading	Distinction
09/2006	08/2009	BSc(Hons)	Nursing (Adult)	United Kingdom (England)	University of Lincoln	2:1

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
10/2015	-	Lead Stroke Research Nurse / HSRC Lead Research Nurse	Imperial College Healthcare NHS Trust	100
05/2014	10/2015	Stroke Research Nurse	Imperial College Healthcare NHS Trust	100
04/2013	04/2014	Stroke Research Nurse	University College London Hospitals NHS Foundation Trust	100
08/2011	04/2013	Rheumatology Infusion Nurse	United Lincolnshire Hospitals NHS Trust	0
11/2009	07/2011	Staff Nurse	United Lincolnshire Hospitals NHS Trust	0
01/2009	06/2009	Student Consultant on Teaching	Centre for Educational Research and Development (CERD), University of Lincoln	20

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
NIHR303439	HEE/NIHR ICA Pre-doctoral Clinical and Practitioner Academic Fellowship 2023	78993	National Institutes for Health and Care Research	12	01/09/2023	31/08/2024

Publication Record
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Publications
Coutts, SB., Ankolekar, S., Appireddy, R., Arenillas, JF., Assis, Z., Bailey, P., Barber, PA., Bazan, R., Buck, BH.,

Publication Record

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Publications
Butcher, KS., Camden, MC., Campbell, B., Casaubon, LK., Catanese, L., Chatterjee, K., Choi, PMC., Clarke, B., Dowlatshahi, D., Ferrari, J., Field, TS., Ganesh, A., Ghia, D., Goyal, M., Greisenegger, S., Halse, O., Horn, M., Hunter, G., Imoukhuede, O., Kelly, PJ., Kennedy, J., Kenney, C., Kleinig, TJ., Krishnan, K., Lima, F., Mandzia, JL., Marko, M., Martins, SO., Medvedev, G., Menon, BK., Mishra, SM., Molina, C., Moussaddy, A., Muir, KW., Parsons, MW., Penn, AMW., Pille, A., Pontes-Neto, OM., Roffe, C., Serena, J., Simister, R., Singh, N., Spratt, N., Strbian, D., Tham, CH., Wiggam, MI., Williams, DJ., Willmot, MR., Wu, T., Yu, AYX., Zachariah, G., Zafar, A., Zerna, C. and Hill, MD.. (2024) ‘Tenecteplase versus standard of care for minor ischaemic stroke with proven occlusion (TEMPO-2): a randomised, open label, phase 3 superiority trial.’, *Lancet (London, England)*.
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Horby, PW., Peto, L., Staplin, N., Campbell, M., Pessoa-Amorim, G., Mafham, M., Emberson, JR., Stewart, R., Prudon, B., Uriel, A., Green, CA., Dhasmana, DJ., Malein, F., Majumdar, J., Collini, P., Shurmer, J., Yates, B., Baillie, JK., Buch, MH., Day, J., Faust, SN., Jaki, T., Jeffery, K., Juszczak, E., Knight, M., Lim, WS., Montgomery, A., Mumford, A., Rowan, K., Thwaites, G., Haynes, R. and Landray, MJ.. (2024) ‘Dimethyl fumarate in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial.’, *Nature Communications*, 15(1).
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Al-Shahi Salman, R., Dennis, MS., Sandercock, PAG., Sudlow, CLM., Wardlaw, JM., Whiteley, WN., Murray, GD., Stephen, J., Rodriguez, A., Lewis, S., Werring, DJ. and White, PM.. (2021) ‘Effects of Antiplatelet Therapy After Stroke Caused by Intracerebral Hemorrhage: Extended Follow-up of the RESTART Randomized Clinical Trial.’, *JAMA neurology*, 78(10).
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Publication Record

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(2017) ‘Cluster-Randomized, Crossover Trial of Head Positioning in Acute Stroke.’, *The New England journal of medicine*, 376(25).
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Bernhardt, J., Churilov, L., Ellery, F., Collier, J., Chamberlain, J., Langhorne, P., Lindley, RI., Moodie, M., Dewey, H., Thrift, AG. and Donnan, G.. (2016) ‘Prespecified dose-response analysis for A Very Early Rehabilitation Trial (AVERT).’, *Neurology*, 86(23).
Bernhardt, J., Lindley, RI., Lalor, E., Ellery, F., Chamberlain, J., Van Holsteyn, J., Collier, JM., Dewey, HM., Parsons, B., Moodie, M., Lennon, S., Donnan, GA., Thrift, AG., Churilov, L. and Langhorne, P.. (2015) ‘AVERT2 (a very early rehabilitation trial, a very effective reproductive trigger): retrospective observational analysis of the number of babies born to trial staff.’, *BMJ (Clinical research ed.)*, 351.
Bernhardt, J., Raffelt, A., Churilov, L., Lindley, RI., Speare, S., Ancliffe, J., Katijjahbe, MA., Hameed, S., Lennon, S., McRae, A., Tan, D., Quiney, J., Williamson, HC., Collier, J., Dewey, HM., Donnan, GA., Langhorne, P. and Thrift, AG.. (2015) ‘Exploring threats to generalisability in a large international rehabilitation trial (AVERT).’, *BMJ open*, 5(8).
Bernhardt, J., Raffelt, A., Churilov, L., Lindley, RI., Speare, S., Ancliffe, J., Katijjahbe, MA., Hameed, S., Lennon, S., McRae, A., Tan, D., Quiney, J., Williamson, HC., Collier, J., Dewey, HM., Donnan, GA., Langhorne, P. and Thrift, AG.. (2015) ‘Exploring threats to generalisability in a large international rehabilitation trial (AVERT).’, *BMJ open*, 5(8).
Wilding, PM.. (2012) ‘Cardiovascular disease, statins and vitamin D.’, *British journal of nursing (Mark Allen Publishing)*, 21(4).
Wilding, PM.. (2008) ‘Reflective practice: a learning tool for student nurses.’, *British journal of nursing (Mark Allen Publishing)*, 17(11).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant

ORCID iD 0009-0003-3016-4751

3. Applicant Research Background

Professional Background	Nurse

Do you have current / active professional registration? This registration must be with one of the ICA approved regulatory bodies
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
09G1826E

Please describe your research career to date

My expertise and passion is in stroke research. I have over 10-years' research experience and provide nurse-leadership to the Hyperacute Stroke Research Centre (HSRC), in partnership with world-leading stroke researchers and clinicians. I have developed comprehensive practice and research skills during my career, I have notable outputs, and supervised student-degree projects. Also, I was awarded the NIHR Pre-doctoral Clinical Academic Fellowship and initiated the Imperial Stroke PPI-Group and Chair the Steering Group, to improve stroke research and clinical service development via diverse lived-experience of stroke, with patient, carer, and public input. The DCAF will help me realise my dream of becoming a clinical academic. I am suitable for this award for the reasons below:

Practice/Clinical and Stroke Research Experience

I have 11-years of stroke research nurse experience, eight-years providing nurse-leadership to the Imperial HSRC and stroke research team. I successfully meet research performance criteria for HSRC re-accreditation every year, backed by the Trust CEO and stakeholders.

My research-experience spans feasibility, setup, delivery, follow-up, and close-out of over a dozen interventional phase-II and phase-III randomised-controlled hyperacute and secondary prevention trials, and rehabilitation and observational studies. My role has been to make stroke research a success at every stage: screening, confirming eligibility, informed consenting, administering stroke scales (NIHSS, mRS, Barthel), cognitive measures (MoCA), quality of life and various other scales, and obtaining accurate, high-quality data. For interventional trials, I support the Principal Investigator and delegated co-investigators to confirm eligibility, obtain informed consent in an ethical manner; and support the patient and relatives with study-information and decision-making. Complex studies, related to thrombolysis and thrombectomy research-delivery for example, require leadership and co-ordination of the multi-disciplinary team (MDT), including stroke consultants, neurointerventionists, and nurses.

I prioritise inclusive and individualised approaches to the patient research experience: Partnering with the PI, research team, and MDT, one study achieved 50:50 gender-representation at our site, compared to a 66:34 gender-asymmetry internationally[113]. The stroke research team and HSRC have been positively covered in national newspaper articles.

Research Outputs and Impact

My academic skills, relevant to the clinical academic pathway, include: Being first author in double-blind peer-reviewed journals[133,134]. Wilding[133] has been the most impactful, critically analysing reflection as a learning modality in the clinical environment; cited in PhD dissertations, chapters-in-books, books, and papers. As a student I co-developed the Students Consulting on Teaching project at the University of Lincoln[21,22].

I have authored and co-authored several double-blind reviewed publications, involved in every aspect of the study design and analysis as full papers, conference engagements, networking and sharing, including presentation of data and sharing of PPI-aspects, stroke research leadership, characterising HSRCs, and other stroke research[11,12,24,50,51,101,135-139]. I have contributed to the expanding evidence-base for stroke treatments and Covid-19. Examples include Attest-2, demonstrating tenecteplase is non-inferior and safe compared to

alteplase for clot-busting treatment (thrombolysis); Pragmatic Ischaemic Stroke Thrombectomy Evaluation (PISTE), demonstrating clot-retrieval (thrombectomy) works and subsequently revolutionised stroke treatments; RECOVERY evidenced three proven-treatments. I am indexed in Medline/PubMed as “investigator” for dozens of published work. I also have experience peer-reviewing for journals, including the British Journal of Nursing.

All of my research activities comply with ICH Good Clinical Practice[60,61] and the highest ethical standards in accord with the Declaration of Helsinki[141] and UK Framework for Health and Social Care Research[53].

PPI

I established a 9-member Stroke PPI Group, and chair a 4-member steering group since September 2023. This developed my facilitation and measuring of PPI-impact skills; uncovering insights into the power and application of diverse PPI-voices to transform and improve stroke research projects and clinical service improvement.

Student Supervision

I supervised students for 4-years for Imperial’s Medical Biosciences (BMB) bachelor’s degree. Students complete a project and work placement instead of a dissertation. I supervised those academic projects to successful conclusions; all four students received a first-class BSc. I also continue to assess and mark student project reports for the university.

Professional Experience

My professional experience equips me to develop as a clinical academic: I qualified in 2009, and specialised in ward-based endocrinology and diabetology, followed by a rheumatology biologics-infusion role; assessing patients for therapeutic benefit and adverse reactions, deploying a wide-range of clinical skills, including venipuncture, cannulation, safe delivery of treatments/infusions of monoclonal antibodies, reviewing of blood results, and managing expected adverse reactions and side effects. Clinical research has been my professional experience since 2013.

Leadership Development

My leadership skills have excelled not only through experiential practice and learning over the past 8-years in a leadership role, but expanded through formal education. I completed the MSc Management for Future Leaders and Level-7 Senior Leader Master’s Degree Apprenticeship, the highest professional-leadership qualification. My dissertation further developed my qualitative research skills; I used semi-structured interviews and thematic analysis to investigate: What Makes an Effective Leader in Stroke Research? This involved interviewing principal investigators, research delivery managers, nursing leads at HSRCs, and others involved in stroke research. My research uncovered competencies that should be prioritised in the hiring of leaders, and selection of effective PIs, in order to enhance patient opportunities for research and improvement of research delivery services; inclusive of under-served populations, with equality, diversity, inclusion, and well-being, being part of the research environment[135,139]. It also highlighted areas for development my subsequent PCAF and consequently this application.

This research experience demonstrated that combining clinical practice and academic research is powerful, where the research nurse can see the benefit to the patient and public: frustrations and challenges of everyday practice providing opportunities and motivation to improve research-options for patients and staff. This first-hand experience provides insight into more effective ways of embedding research into clinical stroke pathways, expanding patient access to research options in an inclusive-manner, and offering treatments unavailable outside of research.

Summary

My leadership and professional experience, resilience, and stroke research career demonstrate my potential to lead stroke research as a practitioner academic and nurse leader. I believe my enthusiasm and interests will have a positive and sustained impact on stroke research, the public, the NHS, and building a stroke clinical academic research community as recognised by the Chief Nursing Officer’s Strategic Plan for Research[78].

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?

No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

Impact on Covid-19 for Research, Publications and Progression to Clinical Academic

The pandemic reduced time for producing publications and outputs, delayed the completion of my master's degree and apprenticeship. Which had an impact on the furtherance of my research career.

I was redeployed to Urgent Public Health (UPH) Covid-19 research during the pandemic. This had positive aspects, such as further developing leadership skills site-wide, especially in a crisis situation, building resilience, dealing with uncertainty, learning about platform study designs, and contributing to the UPH Recovery trial. Recovery delivered three effective treatments to the public. However, the pandemic had numerous negative effects for several years afterwards; meaning but for the pandemic, I would have been able to make earlier progress, and evidence greater potential for the clinical academic pathway, than I already have.

The pandemic meant that the part-time two-year Master's degree for my cohort had to be delayed and extended at several junctures, including after being de-deployed back to stroke research during the reset and recovery period, in response to Covid-19 waves and lockdowns. It also meant balancing the academic requirements of the course with delivering both stroke and UPH research at certain points.

The pandemic also impacted my ability to publish and disseminate my work and potential outputs. I pride myself on developing resilience, and have the mindset of taking positives from hardship and finding solutions to overcome barriers, and growing as a person and professional through reflection and responding to challenges. I am grateful to develop these qualities.

Despite these challenges, the PCAF has put me back on track, but is unable to redress years lost to the pandemic. The PCAF has put me in good stead to undertake this Doctoral Clinical Academic Fellowship, to benefit patients and society, and to excel as a clinical academic.

4. Plain English Summary of Research

Background

If you have a blocked blood-vessel or bleed in the brain (stroke), you should be taken to a specialised centre for urgent treatment to help prevent death and disability. Such centres are called Hyperacute Stroke Units (HASUs) and those that specialise in research are accredited NIHR Hyperacute Stroke Research Centres (HSRCs). If eligible, you should be offered participation in a clinical trial, this might be because no established-treatment option is available for you or because the research option is potentially safer and more effective. However, this doesn't always happen. Even at centres that specialise in urgent treatment of strokes and hyperacute research, patients do not always get the option of participating in a research treatment.

Also, if you are a woman, non-White, from a marginalised community or poorer socio-economic background, you are less likely to participate in research treatments. This means you have less treatment-options available to you when needed most to prevent death or disability.

Aim of the Research

My study aims to find out why and what steps we can take to improve inclusive-participation at HSRCs in the United Kingdom.

My study is needed now because the burden of stroke is estimated to increase by 35% by 2050. Currently stroke is the main cause of adult-disability and the third-biggest killer in the UK, costing £26 billion annually. We need more treatment options to help reduce death and disability, and reduce the impact of stroke. Also, equality diversity and inclusion is a national priority that requires urgent action.

Patients and carers will benefit from my study because it will widen-access and improve participation in clinical trials for eligible patients at HSRCs across the UK. Inclusive-participation will enhance reliability of the evidence produced. Society will benefit because clinical trials will successfully finish on time or earlier, informing guidelines and making clinical treatments available sooner. This, in turn, will reduce the human and economic burden of stroke nationwide. My study will improve the stroke service and also contribute to national strategies related to equality, diversity and inclusion, embedding research into clinical practice, improving commercial-study performance, and improving access to clot-removal (thrombectomy).

Design and Methods

To investigate the issues and develop a solution, I will use the Design Council's Double Diamond Framework, its design-thinking principles, and Normalisation Process Theory (NPT). NPT is a framework to understand the implementation, embedding, and integration of interventions in healthcare settings. These frameworks have been used in healthcare to achieve successful outcomes in complex interventions.

I will adopt a mixed-methods approach: I will speak to patients and carers across different demographics who have experienced stroke to understand what the issues and opportunities in emergency stroke research are. I will observe and speak to stroke service colleagues who deliver research services at 4 HSRCs that have agreed to participate. Following that, in partnership with patients and key stakeholders, I will co-design and test the solution at those centres to improve inclusive participation.

Patient and Public Involvement (PPI)

My stroke PPI-group, composed of patients and carers with lived-experience of stroke, helped and continue to shape my study and identify areas of focus, navigate ethical issues, and validate outcomes. For example, members have confirmed the need for inclusive-participation and to find a sustainable-solution for the post-code lottery access to thrombectomy-research. Also, PPI-input has informed how I am thinking about how and when patients/carers should be contacted, and how gender, ethnicity, and culture can bear on receptivity to research.

Dissemination and Impact

I will share my findings with patients, public roadshows, HSRCs, and wider stroke-community. I will also present at national and international stroke conferences and publish in scientific journals.

5. Scientific Abstract

Background: Lack of inclusive-participation in hyperacute and acute stroke research

Stroke is the third-biggest killer in the UK, impacting quality-of-life, morbidity and mortality, with a vast £26bn economic-burden to society. Advances in stroke research mean new treatments like thrombolysis and thrombectomy are now available. But the window for these treatments is astonishingly narrow (4.5 and 6-hours, respectively) and there are limited secondary prevention treatments for stroke.

Nationwide, 13 NIHR Hyperacute Stroke Research Centres (HSRCs) host extensive clinical trial options. But eligible-patients are not always offered a clinical trial, even when this is the only treatment-option available or is potentially safer and more effective. Further, underserved populations - women, ethnic minorities, and lower-socioeconomic persons miss out disproportionately, for example in thrombectomy research. This is problematic, as evidence might not represent the population, leading to less-reliable results raising safety and efficacy concerns. Trials also take longer to recruit, require extensions, and delay evidence that can change treatment paradigms.

Research Question

How can we improve inclusive participation in hyperacute and acute stroke research at HSRCs in the United Kingdom?

Key Aims and Objectives

Key aims are to (1) discover and define the problems and opportunities related to participation, and (2) develop and implement an effective, framework, co-designed with patients and key stakeholders to improve inclusive-participation in hyperacute/acute stroke research. In order to drive forward new treatments for patients and reduce the impact of stroke on society.

Methods

My study will adopt a convergent mixed-methods approach and use the established Design Council's Double Diamond Framework, design thinking principles, and Normalisation Process Theory, to co-design the solution with patients and key stakeholders. Four of 13 HSRCs that have agreed to participate. Qualitative data collection methods will include ethnographic observation, interviews, and focus groups. Quantitatively, I will analyse data from the NIHR Open-data-platform, Sentinel Stroke National Audit Programme, and HSRC annual-accreditation pro-formas; local screening logs; and local databases for participation, performance, and pathways. I will use thematic analysis and statistical methods and integrate these to inform data interpretation.

Timelines for Delivery

Year 1: Discover and define issues and opportunities, continue with PPI-involvement, and engagement with 4-HSRCs. Deliverables: a well-defined problem domain and opportunities, a complete scoping-review, and theoretical framework.

Year 2: Co-design, develop and implement an inclusive-participation framework at 4 HSRCs. Deliverables: an initial framework and results of testing and evaluation.

Year 3: Final inclusive-participation framework, writing-up, and dissemination of the outcomes.

Post-doctoral work will further test the concepts, solutions, and expand and evaluate their implementation and longer-term impact nationwide.

Anticipated Impact and Dissemination

Short-term impact includes a positively-involved Stroke PPI-group and HSRCs. Medium-to-long term, it will contribute to: (1) patient-access to wider stroke treatment-options including thrombectomy; (2) improving HSRCs' clinical practice of inclusive-participation through better-integrating clinical trials and clinical activity; (3) study-sponsors seeing improved results, making the UK more competitive as a destination to deliver research; and (4) NHS and wider society – reduced human and economic burdens, increased commercial-income, realising national strategies.

Dissemination includes presentations and publications at national/international stroke conferences and journals.

6. Detailed Research Plan

What is the problem being addressed?

Stroke is where the brain is affected by either a blood-clot blocking a vessel or bleeding[109]. Stroke is an enormously consequential both to individuals, their families/carers, and society at large. According to NHS England[76], stroke is the main cause of adult disability and third-biggest killer in England; half of stroke survivors are left dependent with long-term disability; with an annual-cost of £26 billion[100]. Consequences can be severe, but also subtle with hidden-impairments affecting quality-of-life and healthy functioning: psychological, cognitive, movement and behavioural burdens can result from stroke, including depression, anxiety, and post-traumatic stress in patients and their carers[71]. Stroke also influences chronic diseases, for example, impaired cognition and vascular dementia are more likely following severe-stroke[39] and support for so-called “minor” stroke is lacking[25].

Three decades ago, stroke-treatments were limited. Advances in stroke research means several effective treatments are now available, but require early-intervention[109]. Early treatment is most effective in treating and avoiding death, disability, and treatment-related complications. Typically, clot-busting treatment (thrombolysis) can only be given within 4.5-hours, and clot-removal (thrombectomy) within 6-hours of onset. Clinical trials are seeking to expand these limited time-windows, and make treatments more effective and safer. These studies are open at Hyperacute Stroke Units (HASUs) that treat stroke patients and Hyperacute Stroke Research Centres (HSRCs) specialising in these clinical trials. However, not every eligible-patient is offered a clinical trial and those underserved miss out disproportionately.

The problem I will address is the lack of inclusion and participation in stroke hyperacute/acute clinical trials. The challenges/issues are:

(1) Eligible patients not automatically offered clinical trial participation.

Ongoing clinical trials aim to improve and expand hypeacute/acute treatment options. But accessing a clinical trial, even at a specialist centre at the urgent time it matters most, is not automatically offered to eligible-patients, even though the study is open to recruitment at that centre[138]. This is a problem because it might be the only treatment option available to the patient or offer a superior/safer treatment. Recent stroke studies addressed previously-identified factors by permitting emergency-consent, have widened-eligibility, and simplified randomisation procedures, telemedicine-options, and tablet-video consent, yet lack of participation still exists[5,11-12].

(2) Underserved populations are under-represented in stroke clinical trials.

Underserved populations are context specific[84] and constitute a broad-category in stroke research participation. Thombectomy research is illustrative:

Gender. Women and men enjoy similar benefits of thrombectomy[33,106,112,]. However, despite being eligible, women are underrepresented in recruitment to thrombectomy clinical trials relative to the burden of disease[15-17]. The gender-disparity can be as high as 66:34[113]. It is unclear why this so. Need for evidence has been highlighted regarding challenges related to gender and also LGBQTI+ communities[131].

Ethnicity and Social Determinants relate to stroke and their outcomes[4,67,73,75,115,125,141]. Ethnic minorities, compared to Whites, suffer stroke 5-years earlier, with 2-to-4-fold increase in death and adverse outcomes[46]. Similarly, wide-disparities in socioeconomic groups exist for stroke: first-stroke occurred 7-years earlier in the lowest SES-group compared to the highest, with higher prevalence of risk-factors like diabetes and disability[9]. Also, non-Whites are underserved in thrombectomy research[114]. Additionally, barriers including language, cultural, and other-aspects have been recognised[52,74,85,110].

Representative and inclusive sampling are important considerations for both treatment and safety

paradigms[2,95]. Underrepresentation applies to both hyperacute and acute-stage secondary prevention studies[20,96], and can lead to erroneous safety and efficacy study-conclusions[125]. This highlights a need for inclusion.

(3) Lack of inclusive participation impacts clinical trials in myriad ways.

Translating clinical trial results into practice is hindered when there are insufficient participants, leading to underpowered, inconclusive or misleading results. Costly-extensions, if granted, are needed to answer the primary research question and delays result[18]. Women, half the population, are underserved in the stroke context, potentially limiting applicability[2,95]. Commercial trials underperforming in the UK may impact future access to novel therapeutics for patients[30,31]. Addressing these issues can improve trial outcomes and patient access.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to service users and health and social care services?

My study is important, timely, and relevant because:

- Stroke is a common cause of death and disability with limited treatments[109], increasing inclusive-participation in stroke clinical trials is needed to drive forward new hyperacute/acute treatments for patients and society, and reduce the social and economic impact of stroke.
- Underserved populations are under-represented in stroke clinical trials and risk undermining relevance and reliability of evidence for the next generation of clinical treatments[2,17,95]. The reasons for this have been described as complex and lacking detailed investigation[15-17].

Why improving inclusive participation to prevent impact of stroke is needed now:

Stroke burden is estimated to increase in the UK by 60% in incidence and 120% prevalence by 2035[66] at a societal-cost of £26 billion annually[100]. According to the Action Plan for Stroke in Europe 2018-2030, stroke is treatable and manageable[89]. Reperfusion therapies like thrombolysis and thrombectomy have expanded treatment-options and increased stroke-survival in the UK[118,126].

The NHS Long Term Plan[76:64-65] aims to increase the number of patients accessing thrombectomy. Currently, even at advanced-HSRCs, patients with medium-vessel occlusions (treatable in a trial like Escape-MeVO) are not always offered this option. My proposed study, by improving inclusive-participation in clinical trials, will contribute to this and six of the ten-top urgent-priorities in Shaping Stroke Research[118] and Demand Signalling[79:18]. These priorities relate to evaluating new and established hyperacute therapies, access to thrombectomy, and preventing stroke-complications.

Importantly, several clinical trials are trying to improve the efficacy and safety of these therapies, and extend the narrow treatment-window. Barriers to patients and underserved exist, even at centres specialising in these treatments, and needs further investigation. So that research-treatments are offered close to the onset of stroke, when reduction of harm is safest and most effective.

Review of Existing Evidence – How does the existing literature support this proposal?

Literature supports this research proposal in three domains: (1) Research participation issues; (2) Lack of interventions to improve inclusive-participation in hyperacute and acute stroke research; and (3) Challenges and potential solutions in hyperacute/acute stroke clinical trials.

(1) Research participation issues:

Variation in consent, recruitment, and retention exists in RCTs[132]. Poor recruitment is a common theme for study discontinuation[64]. One study[18] found 19% of RCTs were discontinued due to under-recruitment, impairing their statistical power. A total of 48,027 enrolled-patients in 2011 were in studies unable to answer their primary research question[18]. Another found that premature RCT-discontinuation has remained the same this past decade[117]. Most reasons for recruitment failure are preventable[10,117].

Stroke context: Elkins and colleagues' (2006) meta-analysis of stroke-RCTs, assessed recruitment efficiency (participants/site/month) and recruitment rate (participants/month)[42]. Both were influenced by eligibility criteria and organisational structure. Broader eligibility-criteria yielded broader-recruitment. Hyperacute stroke RCTs (<6-hours after stroke-onset) had half-the-recruitment compared to studies with >6-hours. Also, a review of RCTs within 12-hours of symptom-onset by Feldman et al. demonstrated a declining efficiency and recruitment in newer trials (2010-2014) compared to older trials (1990-2004)[45].

The 13-HSRCs met more local recruitment-targets in hyperacute and acute secondary prevention trials compared to non-HSRCs[138]. Meaning recruitment was better at these specialised centres.

In summary, patient-participation, in the unique hyperacute-setting, needs further investigation and improvement.

(2) Lack of interventions to improve inclusive-participation in hyperacute and acute stroke research.

A systematic review by Treweek and colleagues highlighted that interventions improving recruitment have plenty of variety but little depth[123]. Two of three high-quality interventions worked: having an open/unblinded study and telephone reminders to postal non-responders. However, no hyperacute or acute stroke centres/studies were included.

A systematic review of qualitative studies on participant-perspective factors impacting recruitment to RCTs in healthcare found: face-to-face discussions, with clear information, is preferred. Patients are influenced by their perceptions of chance of improvement and burden on time, and by what relatives and professionals tell them. Finally, no one-size-fits-all exists for participation and should be individualised. None of the reviewed studies examined underserved populations, nor did any relate to stroke specifically, especially the unique hyperacute-context of HSRCs. Houghton and colleagues suggest further research is needed on "factors that affect black, Asian and minority ethnic involvement in trials . . . [and] under-represented groups, such as the socially disadvantaged, is needed."[58:18]

Similarly, Farrar and colleagues systematically looked at RCTs, from a recruiter's perspective, using a qualitative-evidence synthesis and identified five overlapping-themes: (1) recruiting in a clinical environment; (2) enthusiasm for the study; (3) judgements about whether to approach a patient; (4) communication challenges; and (5) interplay between professional and recruiter roles[44]. This highlights the role of clinical settings to recruitment, in particular the relative lack of high-quality emergency-setting evidence. My study will provide evidence for HSRCs.

Several studies have explored barriers to recruitment using different research approaches. Some have included studies built-into-RCTs, like the QuinteT Recruitment Intervention[36-37,99]. Some used qualitative research to gather data from observation, focus groups, and interviews investigating recruitment and its barriers[41]. Other qualitative research has examined the process of informed consent[41,102,121,132] and clinical trial study-design[59,127], and a simple framework to uncover barriers[139]. Mixed-methods have also been used to explore and successfully overcome barriers in difficult-to-recruit to RCTs[19,106].

It is reassuring participation can be investigated effectively. However, the setting for all those studies is surgical-oncology, a different context to HSRCs. Nor do they specifically focus on underserved groups, aside from scoping reviews, also outside the emergency-context[52,74,101,110].

(3) Challenges and potential solutions in hyperacute/acute stroke clinical trials

Stroke consensus-recommendations include: broader eligibility-criteria; detailed feasibility; simplicity of data-collection regardless of trial-design complexity; using emergency-consent procedures; infrastructure improvements; telemedicine for consent; use of artificial intelligence to identify potential participants; co-enrolment among RCTs; and registries and databases to assess potential numbers of patients[13,14]. Another study recommended: payment; short and illustrated information-sheets; remote follow-ups; financial incentives; and involvement of patient-groups[5].

HSRC context: Simplified data-collection, randomisation, and broad-criteria, have been incorporated into hyperacute research. Recent examples include Attest-2, TICH-3, Maps-2, Escape-MeVO. However, local and national data demonstrate variation in recruitment and inclusivity among HSRCs with no apparent reason. Escape-MeVO for example has 40 inclusive-recruits at one London HSRC (55%-female:45%-male and 40%-BAME:60%-White) versus another comparable London-centre had 4 (75%-male:25%-female and 25%-black:75%-White).

Maps-2 had 139 recruits versus 10 at another. There is limited study into inclusive-participation at HSRCs. My study therefore aims to close this knowledge gap.

The anticipated value and contributions include:

- **Inclusive and wider-participation in stroke clinical trials at HSRCs.**

My study's key outcome will be generalisable across the UK to HASUs that treat stroke patients, and the HSRCs specialising in these clinical trials, making clinical trials accessible to all eligible patients and achieving a more representative sample. In turn increasing efficiency, time-to-target, and speeding up the journey from clinical trial to guideline clinical treatment.

Consequently, contributing to the NIHR's EDI Strategy 2022-2027[86:6] to understand and redesign our systems and processes making them more inclusive. And support organisations to "reduce the inequalities of outcome which result from socio-economic disadvantage" fulfilling the Equality Act 2010 (s.1(1)). Also drawing on untapped potential of frontline research staff to uncover issues and measures of success[29] through co-design of an implementable inclusive-participation solution with key stakeholders and patients at HSRCs.

- **Equipping HSRCs to become more inclusive, effective, and better integration of clinical trials and clinical activity, benefiting eligible stroke patients.**

This supports core goals of Best Research for Best Health[32] and Saving and Improving Lives[55] by improving: patient-centred research, embedding clinical research delivery, making research matter, making research more diverse and relevant to the whole-UK, and strengthening PPI. This will further integration of research as part of direct clinical care[108] and contribute to Levelling Up[56], Life Sciences[27,28], and sustainability strategies[128] of reducing inequalities and improving health.

- **Improving performance of commercially-sponsored trials, making the UK more competitive.**

Hyperacute and acute commercial trials with improved participation will contribute to reversing the UK's underperformance identified in Lord O'Shaughnessy's Review[27,28].

What is the research question / aims and objectives?

Research question: How can we improve inclusive participation in hyperacute and acute stroke research at HSRCs in the United Kingdom?

Key aim: to develop and implement a co-designed framework to improve inclusive participation in hyperacute and acute stroke research at HSRCs, which will drive forward new treatments for patients and reduce the impact of stroke on society.

Objectives:

(1) To review literature and up-to-date evidence and data (ODP and SSNAP databases) and best practice to inform investigation;

(2) To investigate issues and opportunities of inclusive participation among HSRCs;

(3) To develop and co-design a framework to improve inclusive participation in hyperacute/acute stroke research in HSRCs;

(4) To implement the co-designed framework in HSRCs;

(5) To evaluate and test effectiveness of framework;

(6) To revise and propose a final framework.

Project Plan

This project plan has been informed by a scoping literature review I have undertaken, in consultation with key stakeholders, PPI-input, and previous qualitative research I conducted, which was undertaken to explore leadership at HSRCs[135.139] as well as national HSRC performance[138].

- **Study Design**

I will use the Double Diamond Framework for Innovation[33,34] used in many areas of research[103] including healthcare-research[3,98], for my study design. It is a problem-solving approach that emphasises empathy, ideation, and experimentation. It relies on understanding users' needs and testing ideas for validity, rather than relying on assumptions or conventions. Its 4-phases follow:

The Project

Phase 1 – Discover and Define Problems and Opportunities (months 1-15):

This phase will identify and define issues and opportunities related to inclusive-participation at HSRCs to gain deeper insight and understanding. This will involve observing, spending time with, and speaking to those affected, and gathering and analysing data and performance metrics from official sources related to infrastructure and participation. Specifically, I will discover/define opportunities and issues through design-thinking tools such as patient journey-mapping, root-cause analysis, multi-perspective problem framing; alongside the fieldwork of observation and interviewing patients, carers, and professionals.

Phase 2 – Develop and Deliver co-designed Solution (months 16-28):

Phase 2 will involve developing an initial inclusive-participation framework for HSRCs, co-designed with patients and service-stakeholders. It will also involve agreeing assessment criteria for success and impact-measurement for testing/evaluation and refining and improving development and implementation of interventions.

Co-design is underpinned by design thinking principles (being human-centred: putting people first, communicating visually and inclusively, collaborating and co-creating, and iterating)[33,34]. Co-design approaches have been readily-tailored when required in sensitive or complex circumstances[47]. This co-design process will involve key HSRC-stakeholders, diverse patient-representation, and PPI-input. The co-design process will be one group composed of patients and professionals from 4 HSRCs and data drawn from those sites, with 5 planned workshops.

This group, based on phase-1 research-findings, will select and implement solutions at the 4-HSRCs. Ultimately, there will be generalisable solutions, some applicable-to-all HSRCs, and some sub-solutions for specific centres. These complex interventions will require addressing wider system-interactions involving HSRCs[114]. Consequently, Normalisation Process Theory will inform the development, implementation, and assessment of the interventions. NPT is a well-established and widely-used theory, looking at how interventions become routine and normal through four core-mechanisms (coherence, cognitive participation, collective action, and reflexive monitoring)[69,70]. This will help understanding how this complex-intervention will be "normalised", by providing a framework to interpret data in phase-one and implement/assess in phase-two; exploring the process and structural-problems to identify barriers and facilitators, for successful integration and sustainability.

Methods of Data Collection and Analysis

My study will adopt a convergent mixed-methods approach, combining qualitative and quantitative methods. This allows a more comprehensive analysis of the research problem, through merging and integrating these data[23:83]. There is an abundance of live and historical HSRC quantitative-data on performance and site/HSRC-infrastructure, which can be integrated with qualitative data to offer greater insight[105,138].

- Qualitatively:

(a) Non-participant ethnographic observation - at 4-HSRCs to explore real-time day-to-day running of both clinical and research elements of hyperacute and acute stroke and how patients and service interact, including screening and informed-consent processes.

(b) Interviews - will be conducted with service stakeholders, from frontline research team nurses/practitioners, principal and co-investigators, clinicians, research managers, and others identified through stakeholder and infrastructure mapping. Patients, including those under-served in this context, and those (a) offered research but declined, (b) research offered and participated, and (c) not offered, will also be interviewed. Patients will be recruited from patient groups treated at the 4 HSRCs.

(c) Focus Group/Workshops – 5 will be utilised for the development, implementation, and testing phase.

- Quantitatively: collecting relevant data from the NIHR Open-Data-Platform; Sentinel-Stroke-National-Audit-Programme; NIHR HSRC Pro-formas for annual accreditation; local screening-logs; and local databases. Examples of relevant data include: recruitment, time-to-target, number of research team-members, proportion of consultants GCP-trained and delegated to studies, including commercially-funded, out-of-hours research coverage, etc. National HSRC-data will also be collected for context. Additionally, metrics identified during the qualitative-component will also be analysed, for a fuller picture[23].

Methods of data analysis: Interviews will be analysed using thematic analysis following Braun and Clark[6-8] in a trustworthy[90] and transparent way using reflexivity[94]. NVivo, a widely-accepted tool, will be used for coding themes[48]. NPT will also be used to inform the themes. I will use thematic analysis and statistical methods and integrate both qualitative and quantitative data to inform data interpretation.

Sample size and justification

My purposive sampling strategy differs for each group. The upper-limit is to ensure I can answer my research question and be inclusive of diverse-participants, until data saturation:

- **Patients and carers**

Sample: up to 30-participants or until data saturation.

Patients and carers will be recruited from the 4-of-13 HSRCs nationwide, with advice and support from the NIHR BRC Imperial Patient Experience Research Centre and my PPI-group to ensure diverse-representation. PPI-involvement suggests optimal timing for interviewing patients and carers is after the hyperacute/acute period, several weeks-to-months after.

- **Service – HSRCs**

Sample: up to 30-participants from the 4-HSRCs or until data saturation.

I want to understand the problem and opportunities from the service-perspective. Meyer and Mayrhofer suggest sampling strategy be based on prior knowledge of the population and designed to answer the research question[72:280-281]. In this case, the total “population” service-wise is known. It includes: stroke consultants who act as PIs and Co-Is and lead thrombolysis/thrombectomy calls, interventional-neuroradiology specialists, research nurses/practitioners plus management/leadership; preferably via one-to-one interviews or alternatively small focus-groups. Also, qualitative research seeks information-rich and small samples by comparison to statistical-testing in quantitative research[97:1173].

- **Service-Provider HSRCs: 4.**

Why HSRCs?

Only HASUs can treat hyperacute stroke and have demonstrated efficacy[1]. A subset are HSRC-accredited if they meet specific-minimums in terms of infrastructure, staffing, out-of-hours coverage, and advanced-imaging and intervention techniques, capable of conducting the full-range of hyperacute/acute research. HSRCs outperform

non-HSRCs in meeting recruitment metrics[105,138].

Four HSRCs chosen for high-volume and diversity of patients

Using thrombectomy as an example, a postcode-lottery exists in England. Ten-percent of eligible patients receive thrombectomy in London versus 0-3% elsewhere, primarily due to lack of 24/7-service[119,121]. London performs 31% of all UK-thrombectomies, other regions range from 1-14%, and only 20% of centres have 24/7 access[110]. Patients cannot access a thrombectomy-research option if clinical services are unavailable. London has avoided this by how 24/7 thrombectomy-services operate. Furthermore, London HSRCs serve the most diverse-population in the country. Ethnicity, as an example: Asian-20.7%, Black-13.5%, Mixed-5.7%, White-36.8%, White-other-17%, Other-6.3%[93]. Nottingham also has 24/7 thrombectomy and diverse population[92]. Three-London and one-Nottingham HSRC have agreed to support my study. Moreover, these centres serve a diverse (ethnicity and socio-economic) and high-volume of patients clinically, aiding my investigation and for co-designed development of interventions to have maximum impact.

Selection and exclusion criteria

Patients and carers affected by stroke, including those never in research, previously offered, or those who have been involved in research.

OR

Stakeholders in delivering, leading, or part of service-ecosystem related to research.

Exclusion: Those not meeting the selection criteria

Dissemination, Outputs, and anticipated Impact

How will you inform and engage service users, services, and the wider population about your work?

Dissemination of my outputs will be guided by How to Disseminate Your Research[82] and VICTOR tool[57]:

National-level HASUs/HSRCs

I will present at the annual-HSRC meetings for the 13-HSRCs (approximately 40-attendees), where local and national NIHR and HSRC leadership meet and share past performance and future plans.

National and International Stroke Conferences

Delivering presentations, workshops, posters and abstracts at broadly-attended stroke conferences. Including the European Stroke Organisation Conference, World Stroke Conference, International Stroke Conference, and others. Potential for broad-impact EDI-impact on underserved-patient inclusion, and research time-to-target will be of significant interest. This is evidenced by outputs for my PPI and HSRC-activities so far[135-139] .

Local, National, Communications

Shared within my Trust, region, national, through: NIHR RDN-newsletters, internal communications, and NIHR social-media. Engaging PPI-groups across the country. If there is media interest, TV-interviews and print, for UK and worldwide coverage.

Publications in Scientific Journals

I aim to publish at least 3 high-quality journal papers, focusing on the co-design process, PPI, and improving inclusive-participation to inform evidence-based practice, reaching: policymakers, professionals, and research community. Including EDI and PPI-specific findings, following best publication practices[91].

Anticipated impact

My research-outcome will streamline research-integration into hyperacute clinical-pathways, addressing urgent health issues and influencing practice, guidelines, and future work. This will pave the way for post-doctoral work that will further test the concepts, solutions, and expand and evaluate their implementation and longer-term impact nationwide.

What further funding or support will be required if this research is successful (e.g., From NIHR, other Government departments, charity, or industry)?

As a developing clinical academic I would like to expand my study and therefore will apply for the NIHR Clinical Academic Advanced Fellowship and other appropriate post-doctoral funding-streams based on consultation with my supervisors.

What do you think the impact of your research will be and for whom?

Impact of Research

Short-term impact on my PPI-group: They are keen to give-back and have found satisfaction in helping service-improvement. Likely so for patients and the myriad professionals in my study. High engagement-levels and enthusiasm was evident for my recent qualitative study[135,139]. So too for the presentation about my proposed study at the national annual HSRC meeting in March-2024, where all 13-centres attended, demonstrating immense interest and support.

Below are both medium and longer-term impacts.

Impact 1: Patient access to wider clinical trial treatment options

It is crucial to ensure all eligible patients have access to stroke clinical trials. This is currently not the case even at HSRCs. My study's focus on increasing inclusive-participation in hyperacute/acute stroke research at HSRCs can have significant impact, potentially providing more treatment options for patients.

Using a thrombectomy example: Patients with a blocked medium-vessel (an artery in the brain) within 12-hours are currently ineligible to have it removed. Removing it works in theory, but lacks evidence. HSRCs support the international, multicentre, randomised-controlled trial Escape-MeVO (CPMS-52652). Patients can be randomised to having the clot-removed, instead of no treatment.

This is important because being offered a research-treatment option is an important end in itself, while helping achieve several national strategies related to thrombectomy[76], as well commercial and non-commercial research[30,31].

Patients will have better access to several clinical trials:

- NoVEL (CPMS-TBC) – Can we enhance thrombectomy by infusion of an oxygen-donating emulsion peri-procedure?
- Actisave (CPMS-49946) – Can we enhance efficacy of thrombolysis (with or without thrombectomy) using antiplatelets?

Other examples include: expanding clot-busting treatment for patients on blood-thinners (TBC-CPMS-TBC), reducing impact of bleeds (TICH-3-CPMS-50316), preventing pneumonia (MAPS-2-CPMS-50728), stem cells (Revive-CPMS-55472) and acute-stage secondary prevention studies: Librexia Stroke (CPMS-55326); Oceanic Afina (CPMS-58376), and more.

Impact 2: Improved clinical practice of inclusive participation at HSRCs

My framework will enhance practices and clinical-integration of inclusive participation in stroke research. Benefiting frontline staff and stroke stakeholders's wellbeing. Wellbeing, as measured by staff motivational engagement, increases through involvement in research, yielding better patient feedback[63]. Centres with greater numbers of interventional-study participants have lower mortality rates[62].

It will also impact the future direction of HSRCs. One, because there is limited data surrounding HSRCs despite their importance. Aside from one paper related to one trial[105] (and my own preliminary research looking at HSRC participation and performance metrics[138] and stroke leadership effectiveness requirements[135,139], and setting up of a Stroke PPI-group[137]. Two, because it will influence the evolution of HSRC criteria: re-evaluation of its key classifications of hyperacute, complex, non-complex, etc. in the era of thrombectomy and perfusion

imaging. Existing HSRCs will better retain accreditation and achieve targets (usually 15 complex-hypearacute and 50 non-complex hyperacute recruits-per-year). Finally, it is relevant to HASUs: centres lacking infrastructure and high-performance criteria of HSRCs, but also conduct hyperacute/acute research. The inclusion-framework can apply to other emergency research, including treatment of heart attacks with PCI (heart-version of thrombectomy); and traumatic injury.

Impact 3: Study sponsors seeing better results

My framework will improve time-to-target results and completion of clinical trials, with better population-representation. Answering their research questions more efficiently, without extensions. In turn, leading to better and rapid evidence to inform clinical treatments and guidelines. Additionally, commercial-performance and income will increase, making the UK more competitive.

Impact 4: NHS – reduced economic burden and increased commercial-income

My study will help contribute to delivering current national policies and strategies. For example, via a clinical trial, accessing thrombectomy, which is a key NHS Long Term Plan priority and stroke-specific organisations[76,79,118]. Better participation in commercially-funded studies, giving patients access to cutting-edge novel therapeutics and enhancing commercial study delivery to make the UK the best place to do research[30,31]. It will help embed EDI-practices, improve patients' lives, a key part of Best Research for Best Health[32], as well EDI[86] goals and overlapping strategies. And help embed a research culture[87] and make research part of clinical care[108].

Project Management

I will use the following structure to manage day-to-day running of the project:

- Adhering and ensuring milestones as per the Gantt Chart are achieved and delivered on-time;
- Securing ethical-approval, and good-governance throughout;
- Monitoring and ensuring inclusive-recruitment of participants is on-target;
- Leading and ensuring diverse PPI-input at every stage and monitoring the same;
- Managing and chairing meetings with stakeholders, as set out in the project plan.

Imperial College London will oversee the financial-aspects of the award. We will book courses and conferences early for early-bird rates and achieve value for money.

I will meet with my academic supervisors: Dr. Bentley every week, Professor Robert and Dr. Webb, monthly; providing mentorship, trouble-shooting, and guidance throughout. And practice supervisors, Dr. Jones, monthly, and Professor Ruth Endacott, every 3-6-months, for professional and clinical supervision. I will meet with my project advisory group: composed of a design expert, as well as national HSRC leadership, and a PPI expert, every 6-months.

Ethics

Potential ethical issues for the patient and carer include vulnerability, confidentiality, this is how I intend to address these:

Type of Approvals Required

My research will require HRA and NHS approvals, and Research Governance and Integrity Team support at Imperial College London. It will also require Confirmation of Capability and Capacity (CCC) from Imperial College Healthcare NHS Trust and three other HSRCs at other Trusts, where I will observe and access stroke stakeholders sites on-site.

Highest Ethical Standards and Legal Compliance

My study will be conducted in accordance with the Declaration of Helsinki[141], UK Framework for Health and Social Care Research[53] and ICH Good Clinical Practice[60,61]. Confidentiality will be paramount. All data collected will be stored securely in compliance with the General Data Protection Regulation and Data Protection Act 2018, and follow local archiving SOPs after project completion.

Informed Consent for Patients and Professionals

Research into stroke often includes unexpected events and ethical challenges[65] (Kevdzija, 2024). This will be addressed as follows: Participants will be offered an ethically-approved information sheet, explaining the study, their rights, and support. Making clear participation is optional and can withdraw any time, that any decision will be

without detriment to them[60,61]. The information sheets will be written in plain language and informed by PPI-input. Only participants who have offered their informed consent will go forward. Patients lacking capacity or unable to communicate will have their story told via consenting relative or carer.

Ethical Issues for Patients and Researcher

Psychological and sometimes physical risk exists researching sensitive topics[35]. However, observation and interviewing is generally low-risk. Emotive-aspects of the patient experience for both patient and carer, and myself, are likely to be encountered. Speaking to patients and carers will only occur after informed consent (of participant or of relative if patient lacks capacity) and an empathetic approach that facilitates a safe experience, and that allows for breaks and flexibility in response to individual circumstances. Acknowledging my being an "insider" on this research and known at these centres is an enormous advantage, but also requires transparency[123]. This will mitigate against potential harms or undesirable experiences.

Another ethical dimension is patients/carers requiring input outside of research. For example, based on PPI-input and other researchers' experience, many stroke patients mention their side effects due likely to statins, and ask about this class of medicines. This occurs when they encounter a healthcare professional or researcher, and is widely-prescribed for stroke secondary prevention[134,109]. In such a case I will acknowledge my scope of practice and role as researcher, and therefore sign-post or connect the patient to relevant services/point of contacts. Similarly, complaints and discussions about rights and the service will be connected with PALS.

Success Criteria

Success measures:

- Recruitment of patients and service stakeholder according to plan;
- Publications and conference presentations – widely-shared to benefit patients, carers, and professionals;
- Completion of substantial and wide-ranging training and development programme;
- Equipped with relevant research- and professional-skills to be a leading clinical academic;
- Completed fellowship with impactful outputs, and transition to a clinical academic;
- Awarded a PhD by research for improving inclusive participation in stroke research.

Potential risks and mitigations:

- The ambitiousness of my study, which is required to solve this urgent problem. Early involvement, input and full-support from HSRCs nationwide will facilitate full-realisation of this project.
- Insufficient participation. Risk would be not recruiting enough participants nor diverse set of patients. Therefore I will monitor recruitment and diversity, forming action plans and working closely with my PPI-group, PERC, and support from Egality Health to ensure diverse participation[38].
- Accommodation of patient preferences to facilitate participation. Issues related to travel or accessibility due to logistics or patient/carer preference can be overcome by use of technology, such as MS Teams and Zoom; when face-to-face is not possible or preferred.
- Navigating Ethical Issues. PPI-input will ensure I can deal with ethical issues optimally[81]. Also, in securing a diverse strata of patients and carers. I will document and monitor diversity of both PPI groups, participants, and co-design process, and formulate action-plans if required[49,83].

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

Who has been involved and why

My stroke PPI-group, composed of patients and carers with lived-experience of stroke, helped shape my study at every stage. The broader 9-member Stroke PPI-Group I co-founded for research and clinical service development at the Trust[137], and 4-member steering group that I chair, have been pivotal in understand how to approach my study's development and what to prioritise. I gained valuable and detailed input from 5 patients/carers (3 patients, 2 carers – 4 female 1 male - 2 BAME 3 White). The role they played was to share their experiences and perceptions of research and inclusion. I involved these patients/carers because, (1) they have lived-experience of what I want to study (hyperacute/acute participation) and (2) the diversity of the patients for better insight (inclusive participation). Including carers and relatives, as well as patients, was important, as stroke clinical trials are discussed when patients arrive for hyperacute treatment and lack capacity, meaning a relative might consent or be consulted as part of emergency professional-consent. Or they be part of the decision-making process when capacity is intact. They also have expertise navigating the system and interacting with clinical and research services as part of the hyperacute pathway.

What was influenced/changed

My initial idea was to look at service leadership, but PPI-input led me to the idea of having patients and carers with lived experience be part of the holistic co-design process in partnership with the HSRC services, to enhance relevance, add diverse-viewpoints, and orient professionals to patient/carer-responsive solutions.

The term post-code lottery for thrombectomy research also emerged as a new way to think about an issue, which still exists in London (in contrast to 24/7 clinical thrombectomy being available pan-London). This term mirrors the clinical issues outside of London, but for research.

Role of gender, ethnicity, and culture was made relevant in a new way on receptivity to research: that patients and relatives might perceive who it is that is discussing research with them and to incorporate looking at this into my observation-stage research, and whether diversity of professionals has an influence on research participation of patients.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

How PPI will be involved: Patients, carers, and public involvement is crucial for the success of my study, and will continue to be actively involved in every aspect of the research cycle. My priorities are collecting and understanding diverse views, PPI-input will be used, for example, to gauge relevance to patients for the semi-structured interview question-design; to minimise burdening patients when interviewing; to ensure barriers to participation are identified and overcome; to check coded themes from thematic analysis against lived experiences; to seek wide-accessibility and comprehensibility of information sheets and outputs to be disseminated to patients affected by stroke. Most importantly, uncovering insights, opportunities, and improvements that only diverse and holistic perspectives can provide.

How I plan to manage, co-ordinate and capture PPI-input: I will manage and co-ordinate regular quarterly-meetings, to obtain the expertise of the 9-member Stroke PPI-Group that I setup and run, and the steering group that I chair. My goal is to ensure public involvement with lived-experience of stroke that represent this diverse population, especially under-served communities disproportionately affected by stroke, always prioritising EDI. For instance, Marney Williams is a stroke survivor, PPI-specialist, member of my PPI steering group, and one of three NIHR Stroke National Specialty Group members. Marney will continue to coach and help me problem-solve and avoid pitfalls in ensuring meaningful, diverse, PPI-partnership[49,83]. Impact, on the research, me, and the patients will be captured using a well-known template and adapted as needed. I will also document relevant facts, including diversity characteristics.

How I will support and enable service users and carers: PPI-time and travel expenses have been fully-costed. The training offered to PPI-Group members will be the Imperial NIHR BRC Public Involvement Training[85]. If further training is required, for example, a medical or research aspect, bespoke training will be organised from local expertise supportive of this study and in consultation with PPI-specialists.

Why is the approach appropriate? Because it will make my research more inclusive and effective, especially because my study examines patients-accessing stroke research options at hyperacute centres (HASUs/HSRCs), where patient views are of central importance. It is also an urgent national priority[84-88].

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A.

8. Training & Development and Research Support

Proposed training and development programme

My Pre-doctoral Clinical Academic Fellowship (PCAF) developed my knowledge and research-skills in systematic reviews, advanced qualitative and quantitative research methodologies, SPSS and data management principles, Experience-based Co-Design (EBCD) and Patient and Public Involvement (PPI) training. In particular, incorporating PPI-training to effectively set-up and run a PPI-group for research and service-delivery input and improvement. Furthermore, I put other skills into practice via conducting quantitative research and presenting my work in different forums, stroke and leadership conferences, the national Hyperacute Stroke Research Centre (HSRC) annual-meeting, and local-stakeholder divisional meetings. The feedback and experience, with input and consultation with my supervisors, helped me develop the following research and professional development plan. I used the Vitae Researcher Development Framework[129] to identify further development to achieve the PhD and professional development to transition to a clinical academic.

Formal study

I will complete the following modules at Imperial College London's award-winning Graduate School, and meet early-stage and late-stage review requirements:

Year 1: Becoming an effective researcher; power of diversity and inclusion in research; enhancing wellbeing for doctoral researchers; academic resilience; time management for your doctorate; plagiarism awareness; science, research and integrity.

Year 2: A scientific approach to research communication; conferences and seminars; advanced presentations.

Year 3: Critical thinking for communication; planning and preparing for your thesis and viva.

Academic and Professional Training

A1 Knowledge Base – Research Methods: Theoretical Knowledge and Practical Application

Mixed Methods in Health Research - University of Oxford.

This builds on my completion of the Advanced Qualitative Methods and Quantitative learning during my PCAF. My proposed study uses mixed-methods for greater insight through a more comprehensive analysis of the problem. This course covers the philosophical underpinnings, skills, and how to design, analyse, assess, and critically-appraise mixed-methods studies, and the practicalities involved.

Behavioural Science and Complex Interventions - University of Oxford.

This course explores the psychological aspects of behaviour change in healthcare professionals and patients. This is an important part of my PhD, including complex-interventions and interpretation. It will also be useful in my professional life as a clinical academic.

Randomised Controlled Trials - University of Oxford.

This course explores RCTs from start-to-finish. My PhD is mixed-methods and does not envision an RCT. However, at the completion of this DCAF, I aim to be a fully-equipped clinical academic and have the education and insight to conduct research of all types, and be credible and confident when my post-doctoral work explores inclusive participation through an RCT-design.

Introduction to NVivo for Qualitative and Mixed-Methods Analysis - University of Surrey.

This online two-day course comprehensively covers the application of NVivo, which will aid me with the thematic analysis component of my PhD.

A3 Creativity – Innovation and D1 Working With Others - Influence and Leadership

Systems Thinking Approach to Change Management - University of Oxford.

This 3-day course explores why so many complex-change programmes fail, and how to use systems thinking to fully appreciate complexity, and use pragmatic tools to manage complex-change effectively. This will be important for my PhD as well as a clinical academic leader.

Design Tools for Co-Design - Point of Care Foundation.

This 3-hour workshop builds on the EBCD-course I completed as part of my PCAF. Co-design is a key part of my study design. This workshop was devised following feedback for the EBCD-course and offers practical techniques to build groups and co-design.

Foundation in Patient Experience - Point of Care Foundation.

This CPD-certified course is delivered over 10 online-sessions, with 2-hours per-week. Its aims are to develop confidence and skills in those working in patient experience. This will be useful for the patient-involvement, both as participant in my research plan, and PPI-input at every stage.

D3 Engagement and Impact – Public Engagement and Policy

Engagement Masterclasses - Imperial College London.

These 8-masterclasses are for doctoral students wanting to further public engagement. This complements my PPI-training and experience, aimed at broader public engagement. This will be important for on-going public-involvement in dissemination at key stages in my PhD.

Public Policy Analysis - London School of Economics.

This 10-week online-course, requiring 7-10 hours per-week of commitment, will ensure an understanding of the complex factors of social, political, economic, and institutional forces that influence public policy. This course will support my becoming a clinical academic that can influence policy at a national scale to facilitate the improvement of stroke research at all levels. This builds on my management and leadership education and experience.

Professional Development

After discussion with my supervisors, and research leadership at the Trust, I believe my professional development is best served by pairing with research and leadership Trust-wide and nationwide. This is because I would like to be exposed to the “bigger picture” of where research fits into the whole. It will allow me to further expand my network and knowledge. This learning and experience will benefit my ultimate goal of influencing policy as a clinical academic to better enable me to translate my research and collaborations into reality, to benefit patients and wider-society. To this end, I have arranged time with research and executive leadership at my Trust; time with leadership at the newly-arranged NIHR Research Delivery Network (RDN) and the North London Regional RDN (RRDN), shadowing a range of team members in the Research Delivery Directorate and beyond who are involved in all stages of the study life-cycle.

Stroke Placements

I plan to develop my clinical skills and insights through broad-placements at other NIHR HSRCs, including Newcastle, Southampton, Manchester, Birmingham, Stoke on Trent, and Cambridge. As well as a placement in Essen, Germany to see how stroke practice and research occurs internationally.

Conference attendance

B3 Professional and Career Development – Networking and D2 Communication and Dissemination – Communication Methods

I plan to present outputs at every stage of my PhD. In particular, PPI-engagement, the co-design experience, and improving inclusive-participation. Networking will also enhance both my research and professional development as a clinical academic, and secure future collaborations:

Year 1: World Stroke Organisation Conference–October-2025. Year 2: International Stroke Conference–February-2026 and European Stroke Organisation Conference–May-2026. Year 3: European Stroke Organisation Conference–May-2027.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Dr Paul Bentley

Institution	Imperial College London

Position	Senior Lecturer and Consultant Neurologist/Stroke Physician

Description of support provided
Dr Paul Bentley is a consultant neurologist, stroke physician, and senior lecturer at Imperial College London, Imperial College Healthcare NHS Trust, and Clinical Director of the Imperial College Network of Excellence in Rehabilitation Technology. He supervised Pete's PCAF and has extensive experience in stroke treatment and leading research into stroke, including hyperacute and acute stroke research. He has secured funding and developed numerous research studies, including the Gripable study (NCT04475692). Dr Bentley has expertise in quantitative research methods, and has investigated a breadth of topics: acute, clinical, imaging, technology, artificial intelligence and big data, genetics, and meta-analyses. His meta-analysis of Biochemical Determinants of Stroke in PLOS has more than 100 citations. Dr Bentley has supervised 6 PhD students to completion, and is currently supervising 6 PhD students. Dr Bentley will be Pete's primary academic supervisor for this Fellowship. Pete's research interest is stroke, and in particular in improving inclusive participation in hyperacute and acute stroke research and better integration of hyperacute clinical and research pathways, Dr Bentley's familiarity with this area and broad experience, and quantitative methodology expertise underscore his suitability to supervise Pete's PhD. His clinical and research subject-matter and quantitative expertise, is complemented by Professor Glenn Robert's qualitative and co-design expertise. Dr Bentley will work alongside Professor Glenn and Dr Alastair Webb to support Pete through weekly meetings, ensuring input from his other supervisors, and access to the support and expertise needed, both inside and outside of Imperial, to be successful.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Professor Glenn Robert

Institution	King's College London

Position	Healthcare Quality & Innovation Chair, Vice Dean (Research/Impact),

	Head of Division, Methodologies

Description of support provided
Glenn Robert is Professor of Health Care Quality and Innovation, Head of Division of Methodologies, and Vice Dean Research and Impact at the Florence Nightingale Faculty of Nursing at King's College London. Professor Robert has been a Principal Investigator or co-applicant on over 40 externally-funded research grants since 1996, including 17 National Institute for Health Research (NIHR) grants (5 as Principal Investigator). His widely-published research draws on the fields of organisational studies and sociology, incorporating the study of innovations in the organisation and delivery of healthcare services as well as quality-improvement interventions. He has an overarching interest in organisation development and change management that spans all three domains of healthcare research, policy, and practice. Since 2004 Professor Robert has been on developing and testing an innovative 'Experience-based Co-design' (EBCD) approach for improving service quality and developing complex innovations. It has been applied and evaluated in numerous settings in the NHS and internationally. Professor Robert has previously successfully supervised 1 DCAF and 7 other PhD students to completion. He is currently supervising 6 PhD students including 2 DCAFs. Professor Robert will provide academic supervision in partnership with Dr Bentley and Dr Webb. He will support Pete with monthly meetings and provide methodological, co-design, and design thinking expertise as Pete's PhD utilises a mixed methods and co-design framework to address improving inclusive participation in stroke research. As well as supporting Pete to overcome any challenges inherent in undertaking a PhD and complex interventions, and support his transition to a clinical academic.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Dr Alastair Webb

Institution	Imperial College London

Position	Clinical Reader in Stroke Medicine

Description of support provided
Dr Alastair Webb is Clinical Reader in Stroke Medicine at Imperial College London and Consultant Neurologist at Imperial College Healthcare NHS Trust. He has been Chief or Principal Investigator for multiple stroke studies investigating the causes of cerebrovascular disease, including the recently published OxHARP randomised clinical trial, sponsored by the University of Oxford, and has published widely in high-impact journals. Over his career he has been lead or co-applicant in 14 successful research-related grants, including NIHR grants and MRC and Wellcome clinical research fellowships. He was the founder of the European Stroke Education Platform; editor of the neurology section of the Oxford Clinical Guidelines; working group member for the Action Plan for Stroke in Europe 2018-2030; Chaired the European Stroke Organisation's Education Committee and Secondary Stroke Prevention Guidelines, and is currently Chair of the British and Irish Association of Stroke Physicians' Scientific Committee. Before his recent move to Imperial, Dr Webb was the Stroke Specialty Lead for the NIHR CRN Thames Valley and South Midlands. Dr Webb is currently supervising 2 PhD students, and Pete will be his third if successful. Dr Webb will provide academic supervision in partnership with Dr Bentley and Professor Robert, providing stroke specialism expertise, as well as supervision and support on career development, overcoming challenges, understanding academia and broader research systems, understanding study design and successful Grant-applications. Dr Webb's supervision will help ensure this Fellowship is a success and supports Pete on his trajectory toward becoming a clinical academic.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Helen Jones

Institution	Imperial College Healthcare NHS Trust

Position	Lead Nurse for the Clinical Research

Description of support provided
Dr Helen Jones, DHC (Nursing) MSc, BSc (Hons), FHEA, RGN, will act as Pete's practice supervisor. Her experience and expertise will support his practice development, the execution of his professional development plan, and overcoming any challenges that arise. Her mentorship will support the development of his professional skillsets, his journey as a clinical academic, and provide access to opportunities and wider experience. It will also give Pete access to an extensive network. Her unique sector knowledge and experience is broad: she has worked in clinical research for 30 years, amassed a vast amount of experience and skills concerning research delivery and was a member of the NIHR 70@70 Nurse and Midwife Research Leader programme. Currently she is the Lead Nurse for the Clinical Research workforce across Imperial Healthcare NHS Trust and oversees a workforce of approximately 300 post holders. She has previously mentored staff to apply for NIHR and Charity fellowship opportunities. Recently she and her co-applicant were awarded the BMA Foundation Wesleyan Grant to study the "impact of airborne PPE on physical and mental wellbeing". Dr Jones has a national presence in research nursing and has lectured and published on various aspects of the role and workforce. She supports numerous national research committees and organisations including: a. NIHR RRG Advisory Group. b. Contract Management Board for NIHR Centre for Engagement and Dissemination. c. NIHR CRP Programme Board. d. Chair of RCN Research Nurse Subcommittee.

PRACTICE SUPERVISOR 2

Name	Professor Ruth Endacott

Institution	National Institute for Health and Care Research

Position	Director of Nursing and Midwifery at National Institute for Health and care Research

Description of support provided
Professor Ruth Endacott is the NIHR Director of Nursing and Midwifery and holds a PhD in Medical Science. Prior to her current post, she was Professor of Clinical Nursing at both University of Plymouth, UK, and Monash University, Melbourne, Australia. Professor Endacott has held leadership positions as Head of School, Head of Department and Associate Dean (Research) in the UK and Australia. She is Emeritus Professor at Monash University and Trustee/Director at the Intensive Care and National Audit and Research Centre (ICNARC). Professor Endacott has vast clinical and research experience, including with the NIHR Doctoral Clinical Academic Fellowship (DCAF) scheme, and continues to be involved in research and is published widely. Professor Endacott will supervise and support Pete in his professional, practice, and career development, meeting with him every 3-6 months. In addition to seeing how other hyperacute stroke services operate, Pete has identified the goal of achieving insight into national leadership of research and becoming an effective clinical academic, with

the ability to understand and influence local, regional, and national systems, in order to enhance research at all levels so as to improve the lives of patients and society. Professor Endacott will help ensure the successful execution of his professional development plan, overcoming barriers, and making the most of his placements and plan in the context of national leadership and research. Her supervision will help Pete transition to a successful clinical academic.

Collaborations

My PCAF has been a very collaborative enterprise, and this will expand as part of the DCAF.

Collaboration with other NIHR Hyperacute Stroke Research Centres (HSRCs)

Collaborating with national and regional NIHR HSRC leaders is crucial for my study design and support. This nationwide-network unites commercial sponsors, academia, Hyperacute Stroke Units, and HRSCs for stroke research. This collaboration will enhance relationships, research, and my skills in this field.

The following collaborators have extensive clinical and research experience, and will remain an interface between NHS Trusts and Imperial College London. I will conduct my study at Imperial, and three other HSRCs:

- St George's Hospital, supported by HSRC consultant-lead Dr Liqun Zhang, and HSRC research lead Rita Ghatala.
- King's College Hospital, supported by HSRC consultant-lead Dr Yee Mah, and HSRC research lead Maria Consuelo Tibajia.
- Nottingham University Hospitals NHS Trust, supported by HSRC consultant-lead Professor Philip Bath, Nottingham University Hospitals NHS Trust.

Collaboration with National and Regional HSRC Leadership

My continuing collaboration with national and regional HSRC leadership, will give a broader context to where my current role sits, locally, regionally, and nationwide. This will facilitate my journey as a clinical academic and better place me to influence and lead doctoral/post-doctoral research; and improve inclusive participation in stroke research.

Broader collaborations include:

- Placements in 8-HRSCs, which will enhance collaborative working and build relationships for my DCAF and nationwide post-doctoral work.
- Marney Williams is a stroke survivor, PPI-specialist, and one of three NIHR Stroke National Specialty Group Members for Public Involvement. She also sits on the Steering Group that I chair for the Imperial Stroke PPI Group. Her collaboration will continue to be transformative on my PPI-facilitation, skillset-development and access to the Imperial Patient Experience Research Centre. PERC provide excellent community links to those with lived-experience of stroke and under-served communities, as well as training and support.
- Professors Christine Roffe and Chris Price, the chair/lead of the NIHR Hyperacute Stroke Research Centre Oversight Group, and NIHR National Stroke Speciality Lead, respectively, will collaborate to ensure national engagement and support and be part of my project advisory group.
- Drs Dheeraj Kalladka and Joseph Kwan, North-West-London NIHR CRN Stroke Speciality Co-Leads, will collaborate and ensure support for my project and practice development.

Regional, National and International Collaboration

Shadowing leadership at my Trust and the national newly-reorganised Research Delivery Network (RDN), as well as a leading international stroke-centre will enhance perspective and influence, including:

- Trust CEO Professor Tim Orchard, Dr Bob Klaber, Director of Strategy, Research and Innovation, and others for an executive-perspective.
- Dr Kate Jones, Director of Research Delivery at the RDN.

- Dr Sharon Barrett, Chief Operating Officer at the North London Regional RDN.
- Professor Ruth Endacott, NIHR Director of Nursing and Midwifery.
- Professor Roland Veltkamp, at Alfried Krupp Krankenhaus, Germany. A world-leading stroke research investigator, who is also Professor of Neurology and Chair of Stroke Medicine at Imperial College London.

Clinical Academics

Finally, I will continue to collaborate and draw inspiration from the network of clinical academics, specifically the Imperial Post-Graduate Research Forum (PGRF), which supports doctoral students, run by Professor Mary Wells. The Qualitative Research Group offers support for this methodology, and the Behavioural Research in Healthcare eXchange (BRIHX) run by Dr Gabby Judah for behaviour change. This is further supported by the Imperial Clinical Academic Training Office, which offers relevant and high-quality training and engagement for clinical academics. All of this is a resource for practical advice, networking, and collaborative-support during the DCAF, and subsequent nurse-led post-doctoral research that I would like to undertake as a clinical academic and nurse leader.

Host Organisation Support Statement
The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

It is with great pleasure that we write in strong support for Peter Wilding's NIHR Doctoral Clinical and Practitioner Academic Fellowship (DCAF) application.

Pete is one of our senior research nurses and his position as a leader in hyperacute stroke research delivery puts him in an ideal position to transition to a clinical academic role through the support of the NIHR pathway. Pete has 11 years of stroke research nurse experience and leads the Stroke Research Team and Hyperacute Stroke Research Centre (HSRC) since 2015. During his NIHR Pre-Doctoral Clinical Academic Fellowship in 2023, Pete has established a Stroke PPI-group and chairs its Steering Committee

For this Fellowship proposal, Pete wants to investigate, and find a solution, to improve inclusive-participation in hyperacute and acute stroke clinical trials at HSRCs, so that every eligible-patient is offered a clinical trial enrolment option. Imperial is one of 13 accredited NIHR HSRCs in the UK and Pete already has access to this robust stroke research infrastructure. Through membership of this network, Pete has established collaborations to facilitate his research and professional development for the DCAF and subsequent post-doctoral work as a clinical academic. His proposed study is supported by Professor Chris Price (NIHR National Stroke-Speciality Lead), Professor Christine Roffe (Chair NIHR Hyperacute Stroke Research Centre Oversight-Group), and NIHR Stroke National Specialty Group Members for Public Involvement.

Pete's academic supervisors for this fellowship are Dr Paul Bentley and Dr Alastair Webb, both in the Department of Brain Sciences at Imperial College London, and Professor Glenn Robert at King's College London. His practice supervisors are Dr Helen Jones, research workforce lead at ICHT, and Professor Ruth Endacott, director of nursing and midwifery at NIHR.

Dr Bentley's research group hosts doctors and allied health professionals conducting groundbreaking research into stroke prevention, treatment, and rehabilitation. This expertise supervision is complemented by Professor Glenn Robert who is Chair in Healthcare Quality and Innovation and Vice Dean of Research at King's College London, and has previously supervised DCAF fellows. His supervision will provide qualitative/mixed-methods, co-design and design thinking expertise. Dr Webb, Clinical Reader in Stroke Medicine in the Department, will further support Pete as an experienced chief investigator of several large, multi-Centre, stroke studies. Pete's two practice-supervisors will support him with his professional development. Dr Jones is a national leader in research nursing and clinical academic who can offer him exposure to a wide-network of leaders across the UK. Professor

Endacott will give Pete insight and opportunities to understand the research-infrastructure and influence policy at the regional and national levels. The nursing directorate's clinical-academic lead (Professor Mary Wells) will also provide Pete with mentorship and support. He will have the opportunity to engage in a range of nursing and research leadership activities through our Nursing and Midwifery Research Committee, Research Delivery Network meetings and Clinical Academic Careers Committee.

The Fellowship will be hosted within a partnership between Imperial's Department of Brain Sciences and Imperial College Healthcare NHS Trust (ICHT) via the Imperial Academic Health Science Centre (AHSC). The AHSC is committed to providing comprehensive support and development environment for research-focused non-medical clinical academic roles and to developing clinical academic leaders in the healthcare professions, with a focus in "bench to bedside" translational research. We will protect Pete's post during his Fellowship. After completion, we will support Pete's progression through the clinical academic career structure within the Trust, in partnership with Imperial and our NIHR Imperial Biomedical Research Centre (BRC), which has a world-leading Brain Sciences theme.

Pete will be supported by our Clinical Academic Training Office (CATO), a centralised pan-professional academic training hub, which has an excellent track-record of developing trainees of all professions into clinical academic careers. CATO runs a varied programme to support clinical academics, such as multi-professional Masterclasses, a non-medical researcher development course, internal research presentations, research symposia and an annual research conference. It has led to the achievement of NIHR and local Charity funding for several well-established programmes for doctors and for nurses, medicine, allied health professionals, pharmacists and psychologists (NMAHPPs), including post-doctoral fellowships. In addition to CATO, Pete will benefit from a range of professional opportunities for networking and development from our Imperial AHSC, NIHR Imperial BRC, North-West London Applied Research Collaboration (NWL ARC) and ICL Health Professionals Academic Group (HPAG) – as well as a formal PhD-programme at Imperial's award-winning Graduate School.

With an academic base in the Department of Brain Sciences, which has an exemplary track-record in supporting Early Career Researchers (ECR), Pete will have access to clinical academics at all levels. The ECR Committee facilitates early identification of needs, development of peer-networks, mentorship and "best practice" sessions to support personal and professional development. The Department is committed to prioritising equality, diversity and inclusion (EDI) and has recently renewed its Athena Swan Silver Award (2021), alongside other EDI-initiatives such as a Brain Sciences' LGBTQ+ and Allies network. The Department is already supporting several pre-doctoral and doctoral students from nursing and allied health backgrounds, including a current NIHR Doctoral Clinical Academic Fellow, also working in the stroke field.

The Trust's vision is to support nurses to become clinical academic leaders who can generate and apply robust evidence to improve care for patients. In the Trust we have developed a strong research culture and run monthly research seminars for nurses and midwives ('Bringing Research Back to the Floor') as well as a monthly postgraduate research forum, specifically set up to NMAHPs undertaking fellowships to share challenges and ideas and to gain critical feedback and peer support.

In the past year, four out of five nurses, midwives and AHPs (NMAHPs) on NIHR Pre-doctoral Clinical Academic fellowships have subsequently achieved PhD funding, and three doctoral trainees have completed their PhDs. We have supported several clinical research nurses to transition into clinical academic training pathways and believe that they have the skills to flourish as independent researchers.

In summary, we fully support Pete's DCAF-proposal, and his career goal of being an independent nurse researcher leading hyperacute stroke research at Imperial.

David Peterson

Clinical Director. Neurosciences.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Peter Wilding

Employing organisation	Imperial College Healthcare NHS Trust

Grade	7

Starting point on scale	47672

Increment Date	25/10/2024

Justification of Costs
This will cover my salary, which will enable me to be seconded to Imperial College London to undertake the DCAF, and allow the Trust to backfill my post and keep it open for my return in 3 years.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	58250
Geographical Weighting (£)	0
Other allowances (£)	291.25
Superannuation (£)	9424.85
Nat. Insurance (£)	6782.70
Current Annual Costs (£)	**74748.8**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mr Peter Wilding
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: This will cover my salary whilst undertaking the DCAF, and allow backfill to my current post.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	58250	58698	58698	175646
National Insurance	6783	6845	6845	20473
Superannuation	9425	9497	9497	28419
Geographical weighting / Other allowances (if applicable)	292	294	294	880

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	180334	0	0	180334
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel for In-person and Residential Courses
Justification of Cost: These costs are for travel to and from training identified in my Training and Development Programme. Including the following courses: Mixed Methods in Health Research at the University of Oxford; Behavioural Science and Complex Interventions at the University of Oxford; Randomised Controlled Trials Course: A Guide to Design, Conduct, Analysis, Interpretation and Reporting at the University of Oxford; and Systems Thinking Approach to Change Management at the University of Oxford.

	Year 1	Year 2	Year 3	Total Cost (£)
	280	0	0	280

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel for Professional Development 20%FTE - Germany
Justification of Cost: This is for flights to and from Essen, Germany to visit the Alfried Krupp Hospital identified in my Training and Development Programme. The hospital is one of the top-rated institutions in Germany and Europe. The stroke department is run by a world-leading neurologist and leader of stroke research, who also retains his London-based position as Professor of Neurology and Chair of Stroke Medicine Department of Brain Sciences, Faculty of Medicine, at Imperial College London.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	300	0	300

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel for Professional Development 20%FTE - UK
Justification of Cost: This is for travel back and forth for my visits to the other Hyperacute Stroke Research Centres (HSRCs) that are not included as study sites, but are part of my professional and clinical development. They include two-week visits to the following HSRCs: Newcastle; Southampton; Manchester; Birmingham; Stoke on Trent; Cambridge; and Leeds.

	Year 1	Year 2	Year 3	Total Cost (£)
	252	252	256	760

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel for Participants
Justification of Cost: These travel costs are for participants on the study who I will interview. Some of my interviews will be virtual and for those preferring in-person interviews, this budget is to facilitate their transport. This may include patients who have mobility difficulties and cannot easily take public transport, so is costed to include taxis.

	Year 1	Year 2	Year 3	Total Cost (£)
	400	400	400	1200

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Travel for Conferences
Justification of Cost: These costs are for travel for conferences over the three years and identified in the Training and Development Programme.

	Year 1	Year 2	Year 3	Total Cost (£)
	200	200	300	700

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence and Accommodation for Courses
Justification of Cost: These costs are for accommodation and subsistence for courses identified in my Training and Development Programme. Including the following training: Mixed Methods in Health Research at the University of Oxford; Behavioural Science and Complex Interventions at the University of Oxford; Randomised Controlled Trials Course: A Guide to Design, Conduct, Analysis, Interpretation and Reporting at the University of Oxford; and Systems Thinking Approach to Change Management at the University of Oxford.

	Year 1	Year 2	Year 3	Total Cost (£)
	3522	0	0	3522

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Subsistence and Accommodation for Conferences
Justification of Cost: These costs are for accommodation and subsistence for the conferences over the three years and identified in the Training and Development Programme.

	Year 1	Year 2	Year 3	Total Cost (£)
	455	455	455	1365

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence for Study Participants
Justification of Cost: These subsistence costs are for participants on the study who I will interview. Some of my interviews will be virtual and for those preferring in-person interviews, this budget is to cover refreshments and meals for in-person participants, as needed.

	Year 1	Year 2	Year 3	Total Cost (£)
	100	100	100	300

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence and Accommodation for Professional Development 20%FTE -UK
Justification of Cost: This is for subsistence and accommodation for my visits to the other Hyperacute Stroke Research Centres (HSRCs) that are not included as study sites, but are part of my professional and clinical

development. They include two-week visits to the following HSRCs: Newcastle; Southampton; Manchester; Birmingham; Stoke on Trent; Cambridge; and Leeds.

	Year 1	Year 2	Year 3	Total Cost (£)
	5845	5845	5845	17535

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence and accommodation for Professional Development 20%FTE - Germany
Justification of Cost: Subsistence and accommodation for Professional Development 20%FTE -Germany

	Year 1	Year 2	Year 3	Total Cost (£)
	0	6003	0	6003

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference Fees
Justification of Cost: These costs relate to the conferences fees to so that I can attend, present, and network, as part of my Training and Development Programme.

	Year 1	Year 2	Year 3	Total Cost (£)
	260	415	215	890

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Hyperacute Stroke Research Centre (HSRC) - Nottingham - Study Site - Travel
Justification of Cost: One of the Hyperacute Stroke Research Centres (HSRCs) is outside of London. I will go to Nottingham for data collection; observation and interviews. These are the journey costs for this.

	Year 1	Year 2	Year 3	Total Cost (£)
	70	0	0	70

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Hyperacute Stroke Research Centre (HSRC) - Nottingham - Study Site - Subsistence
Justification of Cost: One of the Hyperacute Stroke Research Centres (HSRCs) is outside of London. I will go to Nottingham for data collection; observation and interviews. These are the accommodation and subsistence costs for this.

	Year 1	Year 2	Year 3	Total Cost (£)
	2505	0	0	2505

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	13889	13970	7571	35430

Equipment
No costs have been added

Consumables
No costs have been added

Patient and Public Involvement

Description: Patient and Public Involvement (PPI) Payments
Type of cost: HEI
Please describe fully: Patient and Public Involvement (PPI) Payments

Justification of Cost: Patient and Public Involvement (PPI) is an essential part of my PhD in order to obtain lived experience of stroke from patients, carers, and relatives. I have costed for 9 members and 13 meetings in total. The in-depth nature of the PPI involvement is reflected in the £75 payment per person. I have been advised to add £5 to this for data and WiFi costs for virtual meetings.

	Year 1	Year 2	Year 3	Total Cost (£)
	3120	3120	3120	9360

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	3120	3120	3120	9360

Other Direct Costs

Description: Transcription service
Type of cost: HEI
Please describe fully: Transcription service
Justification of Cost: This is to transcribe interviews and workshops, based on 60 interviews at approximately 90 minutes each, and 5 x 2-hour workshops. The rate is based on £1.30 per minute excluding VAT.

	Year 1	Year 2	Year 3	Total Cost (£)
	3120	3120	3120	9360

Description: Laptop
Type of cost: HEI
Please describe fully: Laptop
Justification of Cost: This is for a laptop so that I conduct my PhD and Professional Development over the three years.

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Phone and SIM for 3 years, Dictaphone and Memory Cards
Type of cost: HEI
Please describe fully: Phone and SIM for 3 years, Dictaphone and Memory Cards
Justification of Cost: This is for: A phone and SIM over the 3 years, so that I can have a phone number that I can share, to communicate with participants and others. A Dictaphone and memory cards so that I can record (for transcription, then reflexive thematic analysis), in-person interviews.

	Year 1	Year 2	Year 3	Total Cost (£)
	425	0	0	425

Description: Dissemination: Poster Printing over 3 years
Type of cost: HEI
Please describe fully: Dissemination: Poster Printing over 3 years
Justification of Cost: Dissemination: Poster Printing for 3 posters each year, over 3 x years.

	Year 1	Year 2	Year 3	Total Cost (£)
	180	180	180	540

Description: NHS Study Set-up Costs (R&D set-up fee for Non-Commercial observational study) for the 4 study sites
Type of cost: HEI
Please describe fully: NHS Study Set-up Costs (R&D set-up fee for Non-Commercial observational study) for the 4 study sites
Justification of Cost: NHS Study Set-up Costs (R&D set-up fee for Non-Commercial observational study) for the 4 study sites.

	Year 1	Year 2	Year 3	Total Cost (£)
	2356	0	0	2356

Description: NHS Close Down Costs for the 4 study sites
Type of cost: HEI
Please describe fully: NHS Close Down Costs for the 4 study sites
Justification of Cost: NHS Close Down Costs for the 4 study sites (Archiving fee - per box - to be paid upon completion of study)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2660	2660

Description: NHS SoECAT costs
Type of cost: HEI
Please describe fully: NHS SoECAT costs
Justification of Cost: NHS costs as reflected in the SoECAT

	Year 1	Year 2	Year 3	Total Cost (£)
	2820	0	0	2820

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	9901	3300	5960	19161

Training And Development

Type of cost: HEI
Description: Short Courses
Item Description: Mixed Methods in Health Research at the University of Oxford
Justification of Cost: This 5-day residential course will advance my knowledge and practical application of mixed methods, which I will use in my PhD. It builds on the Advanced Qualitative Methods and Quantitative learning during my PCAF. This course covers the philosophical underpinnings and how to design, analyse, assess, and critically appraise mixed methods studies, and the practicalities involved.

	Year 1	Year 2	Year 3	Total Cost (£)
	3175	0	0	3175

Type of cost: HEI
Description: Short Courses
Item Description: Behavioural Science and Complex Interventions at the University of Oxford
Justification of Cost: This 5-day residential course explores the psychological aspects of behavioural change of professional and patients, relating to complex interventions in healthcare. This is an important part of my mixed-methods PhD, and will be of special importance for the interpretation and implementation phase.

	Year 1	Year 2	Year 3	Total Cost (£)
	2890	0	0	2890

Type of cost: HEI
Description: Short Courses
Item Description: Randomised Controlled Trials Course: A Guide to Design, Conduct, Analysis, Interpretation and Reporting at the University of Oxford.
Justification of Cost: This 5-day residential course explores the RCT for start-to-finish. My PhD is mixed-methods and does not envision an RCT. However, at the completion of this DCAF, I aim to be a fully-equipped clinical academic and have the education and insight to conduct research of all types, and be credible and confident if post-doctoral work explores inclusive participation through an RCT-design.

	Year 1	Year 2	Year 3	Total Cost (£)
	1550	0	0	1550

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to NVivo for Qualitative and Mixed-methods Analysis at the University of Surrey
Justification of Cost: This online two-day course comprehensively covers use and application of NVivo, which will aid me with the thematic analysis component of my PhD.

	Year 1	Year 2	Year 3	Total Cost (£)
	255	0	0	255

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: Applicant PhD Tuition Fees
Justification of Cost: This is to cover Imperial College London's PhD fees, which for me will be £7,280 per year for an anticipated April 2025 start. Each year will rise with inflation, but the exact figures for years 2 and 3 are not currently available.

	Year 1	Year 2	Year 3	Total Cost (£)
	7280	7280	7280	21840

Type of cost: HEI
Description: Short Courses
Item Description: Design Tools for Co-Design at The Point of Care Foundation
Justification of Cost: This 3-hour workshop builds on the Experience Based Co-design (EBCD) course I completed as part of my PCAF. The workshop was devised on feedback from previous EBCD sessions and offers practical techniques to build groups and co-design. Co-design is an essential part of the Double Diamond Framework that my PhD will follow.

	Year 1	Year 2	Year 3	Total Cost (£)
	180	0	0	180

Type of cost: HEI
Description: Short Courses
Item Description: Foundation in Patient Experience Course at The Point of Care Foundation
Justification of Cost: This course is a more in-depth look at further developing the knowledge, skills, and confidence by learning the tools and methods to integrate patient experience improvement. This will help with patient involvement in the co-design stage of my PhD and in the solution package for improving inclusive participation in stroke research.

	Year 1	Year 2	Year 3	Total Cost (£)
	1130	0	0	1130

Type of cost: HEI
Description: Short Courses
Item Description: Systems Thinking Approach to Change Management at the University of Oxford
Justification of Cost: This 3-day course explores why complex change programmes fail, and how to use systems thinking to fully appreciate complexity, and pragmatic tools to manage complex-change effectively. This will be important for my PhD in terms of the complex intervention. It will also be vital as a leader and clinical academic.

	Year 1	Year 2	Year 3	Total Cost (£)
	1295	0	0	1295

Type of cost: HEI
Description: Short Courses
Item Description: Public Policy Analysis at the London School of Economics
Justification of Cost: This 10-week online course, requiring 7-10 hours per week of commitment, will ensure an understanding of the complex factors of social, political, economic, and institutional forces that influence public policy. This course is vital to my aim of being a clinical academic that can influence policy at a national scale to facilitate improvement of stroke research and all levels. This builds on my management and leadership education and experience, combined with anticipated practice supervision from Professor Ruth Endacott in exploring national structures.

	Year 1	Year 2	Year 3	Total Cost (£)
	2200	0	0	2200

	Year 1	Year 2	Year 3	Total Cost (£)
Training And	19955	7280	7280	34515

Development Total				

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	101467	84073	81062	266602

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: Imperial College of Science, Technology and Medicine
Justification of Cost: Estates and Infrastructure

	Year 1	Year 2	Year 3	Total Cost (£)
	27113	27113	27113	81339

Type of cost: Estates Charges
Name of Higher Education Institute: Imperial College of Science, Technology and Medicine
Justification of Cost: Other Indirects

	Year 1	Year 2	Year 3	Total Cost (£)
	85105	85105	85105	255315

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	112218	112218	112218	336654

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	89774	89774	89774	269322

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	229132	269322	498454
Total Higher Education Institution Costs (100%)s	37470	0	37470
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research	0	0	0

Costs			
Total Research Costs Requested From Funder	266602	269322	535924

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**59800**	**60267**	**60267**	**180334**
Travel-subsistence-and-conference-fees	**11295**	**11390**	**6251**	**28936**
Equipment	**0**	**0**	**0**	**0**
Consumables	**0**	**0**	**0**	**0**
Patient-and-Public-Involvement	**2496**	**2496**	**2496**	**7488**
Other-Direct-Costs	**7921**	**2640**	**4768**	**15329**
Training And Development	**19955**	**7280**	**7280**	**34515**
HEI-Indirect-costs	**89774**	**89774**	**89774**	**269322**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**191241**	**173847**	**170836**	**535924**

NHS Support Costs

No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs

No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)

NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	535924
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	535924

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References-PWilding-DCAF.pdf) is included as an appendix within this file.

(Research Timetable - Research-PlanPWilding-DCAF-Final.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_345365_SoECAT_Improving inclusive participation at Hyperacute Stroke Researc.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Dr
Forename(s)	Paul
Surname	Bentley
Position	Clinical Senior Lecturer

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Professor
Forename(s)	Glenn
Surname	Robert
Position	Chair in Healthcare Quality & Innovation

Supervisor Participant 2

Title	Professor
Forename(s)	Alastair
Surname	Webb
Position	Associate Professor

Practice Supervisor(s)

Practice Support Participant 1

Title	Professor
Forename(s)	Ruth
Surname	Endacott
Position	Director of Nursing & Midwifery

Practice Support Participant 2

Title	Dr
Forename(s)	Helen
Surname	Jones
Position	Lead Nurse for the Clinical Research Workforce

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
Forename(s)	Jennifer
Surname	Landmann
Position	Research Proposal Manager

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Marina
Surname	Botto
Position	Head of Department

Head of Department or Senior Manager 2

Title	Mr
Forename(s)	David
Surname	Peterson
Position	Consultant

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
None.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. References-PWilding-DCAF.pdf
2. Research-PlanPWilding-DCAF-Final.pdf
3. Funder Export_345365_SoECAT_Improving inclusive participation at Hyperacute Stroke Researc.xlsx

Appendix #1: References-PWilding-DCAF.pdf

References:

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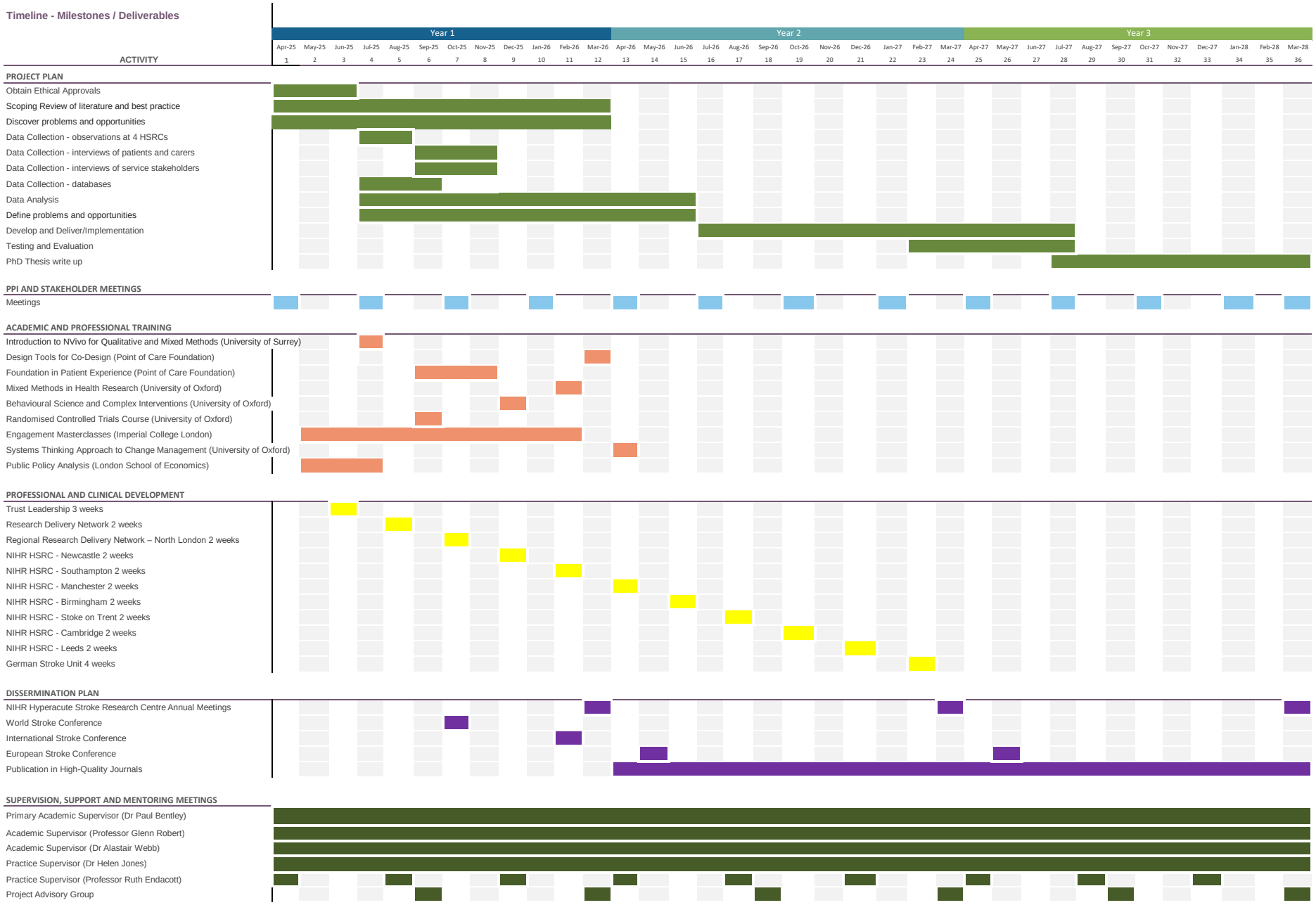
Appendix #2: Research-PlanPWilding-DCAF-Final.pdf

Timeline - Milestones / Deliverables



Appendix #3: Funder Export_345365_SoECAT_Improving inclusive participation at Hyperacute Stroke Researc.xlsx